## Решение

T Условие

S Решение

Русский

A1 0.20 Determine the pressure $p_1$ at the bottom of the vessel. Express your answer in terms of $m$, $g$, and $S$.

Ответ:

$$p = \frac{mg}{S}$$

A2 0.30 Determine the pressure $p(z)$ on the vessel walls as a function of depth $z$. Express the answer in terms of $\rho$, $g$, and $z$.

According to Pascal's law, the pressure of a liquid depends only on the depth of immersion, therefore:

Ответ:

$$p(z) = \rho g z$$

B1 0.50 Write down the equilibrium condition for this layer. The equation may include the pressures $P(z)$ and $P(z + \mathrm{d}z)$, $\mathrm{d}z$, the average pressure on the vessel walls $\sigma(z)$, the mass of the layer $\mathrm{d}m$, as well as $\mu$, $g$, and $R$.

The equilibrium condition projected onto the vertical axis:

Ответ:

$$(P(z + \mathrm{d}z) - P(z))\pi R^2 - \mathrm{d}mg + \mu\sigma(z) \cdot 2\pi R \mathrm{d}z = 0$$

B2 1.20 Show that the bottom pressure $P$ as a function of the fill height $h$ of the granular column can be written as:

$$P = A + Be^{-h/\lambda}.$$

Determine the parameters $A$, $B$, and $\lambda$. Express your answers in terms of the bulk density $\rho$, as well as $g$, $R$, $K$, and $\mu$.

Using the relations $\mathrm{d}m = \rho\pi R^2 \mathrm{d}z$ and $\sigma(z) = KP(z)$, we obtain:

$$\frac{\mathrm{d}P}{\mathrm{d}z} + \frac{2K\mu P(z)}{R} - \rho g = 0$$

Solving the equation and taking into account that $P(0) = 0$, we obtain:

$$P(z) = \frac{\rho g R}{2\mu K}\left(1 - e^{-2\mu K z/R}\right)$$

Let us find the pressure at the bottom by substituting $z = h$ into the expression obtained:

$$P(h) = \frac{\rho g R}{2\mu K}\left(1 - e^{-2\mu K h/R}\right)$$

Hence:

Ответ:

$$A = \frac{\rho g R}{2\mu K}, \quad B = -\frac{\rho g R}{2\mu K}, \quad \lambda = \frac{R}{2\mu K}$$

B3 1.20 Determine the bulk density of the granules $\rho$ and the coefficient $K$. Give the answers to two significant figures. For this part, assume the numerical value of the vessel radius is $R = 50\ \mathrm{cm}$ and the coefficient of friction is $\mu = 0.30$.

For small heights of the grain layer, the dependence of the pressure on the bottom on the height is approximately described by the expression:

$$P(h) \approx \rho g h$$

Let us draw a tangent to the graph at the point (0, 0). The slope of this tangent is:

$$k_1 = 6.5\ \mathrm{kPa/m}$$

Hence:

Ответ:

$$\rho = \frac{k_1}{g} \approx 660\ \frac{kg}{m}$$

For sufficiently large $h$, the pressure on the bottom reaches a steady value. The steady-state pressure is:

$$P_\infty = 26.7\ \mathrm{kPa}$$

Hence:

Ответ:

$$K = \frac{\rho g R}{2\mu P_\infty} \approx 0.20$$

C1 0.10 In the specified approximation, obtain the dependence of the vessel's cross-sectional radius on height $r(z)$. Express the answer in terms of $R$, $\alpha$, and $z$.

Ответ:

$$r = R - \alpha z$$

C2 0.60 Write down the equilibrium condition for the layer. The equation may include the pressures $P(z)$ and $P(z + \mathrm{d}z)$, $\mathrm{d}z$, the average pressure on the vessel walls $\sigma(z)$, the mass of the layer $\mathrm{d}m$, as well as $\mu$, $g$, $r(z)$, $r(z + \mathrm{d}z)$, $\alpha$, and $R$.

Equilibrium condition in the projection on the vertical axis (here it is assumed that $\alpha \ll 1$, therefore $\cos\alpha \approx 1$, $\sin\alpha \approx \alpha$):

Ответ:

$$P(z + \mathrm{d}z)\pi r^2(z + \mathrm{d}z) - P(z)\pi r^2(z) + 2\pi(\mu + \alpha)\sigma(z)r(z)\mathrm{d}z - \mathrm{d}mg = 0$$

C3 0.80 Determine the parameters $A$, $\gamma$, and $\beta$. Express the answer in terms of the bulk density of the granules $\rho$, $g$, $z_{\max}$, $K$, $\mu$, and $\alpha$.

Next, in the equation obtained in the previous step, we will neglect the term $\alpha$ compared to $\mu$.
Note that near the free surface of the granular material, the pressure is zero:

$$P(0) = A(1-\gamma) = 0$$

From this:

Ответ:

$$\gamma = 1$$

The dependence $P(z)$ is described by the following differential equation:

$$\frac{\mathrm{d}(Pr^2)}{\mathrm{d}z} + 2K\mu P(z) r(z) - \rho g r^2(z) = 0$$

Let us introduce the notation $\xi = \dfrac{z}{z_{\max}}$. Substitute the expression for $P(z)$ given in the problem statement into the differential equation, taking into account that $\gamma = 1$:

$$\alpha^2 A z_{\max} \left(-1 + \beta(1-\xi)^{\beta-1}\right)(1-\xi)^2 + 2(K\mu - \alpha) A \alpha z_{\max}(1-\xi)\left(1 - \xi - (1-\xi)^{\beta}\right) - \rho g \alpha^2 z_{\max}^2 (1-\xi)^2 = 0$$

The equation can be satisfied for all $\xi$ only if the coefficients of $(1-\xi)^2$ and $(1-\xi)^{\beta+1}$ vanish.
Therefore:

$$-A\alpha^2 + 2(K\mu - \alpha) A\alpha - \rho g \alpha^2 z_{\max} = 0$$

Hence:

Ответ:

$$A = \frac{\rho g \alpha z_{\max}}{2K\mu - 3\alpha}$$

Also, the following holds:

$$\alpha^2 A\beta - 2(K\mu - \alpha) A\alpha = 0$$

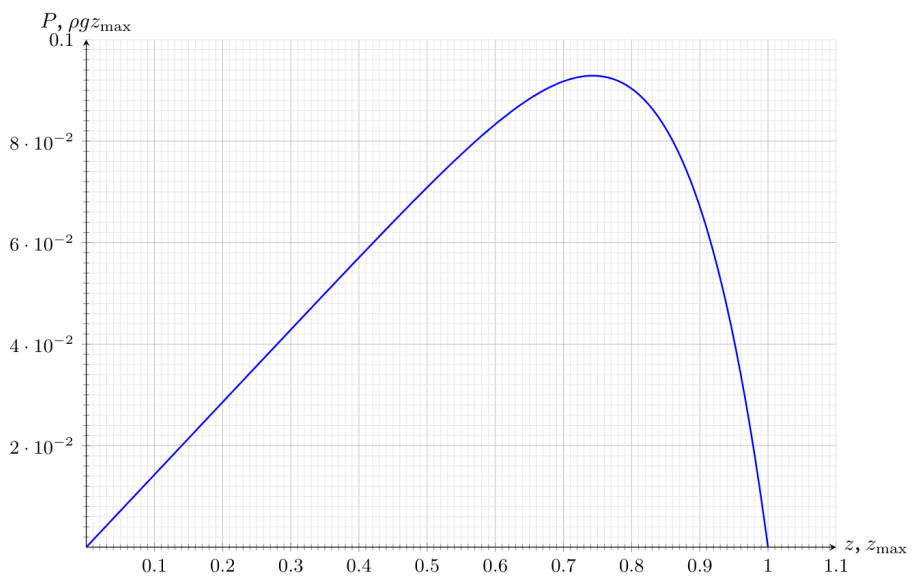
From this:

Ответ:

$$\beta = \frac{2(K\mu - \alpha)}{\alpha}$$

C4 0.80 Sketch a qualitative graph of $P(z)$ and determine the ratio of the maximum vertical pressure $P_{\max}$ to $\rho g z_{\max}$ for $\alpha = 0.2K\mu$. Provide the answer to two significant figures.

Ответ:




For $\alpha = 0.2K\mu$, let us perform the calculations:

$$A \approx 0.143 \rho g z_{\max}, \quad \beta = 8$$

The dependence $P(z)$ has a maximum when $1 - \xi - (1-\xi)^8 \to \max$, that is:

$$8(1-\xi)^7 = 1, \quad z = (1 - 8^{-\frac{1}{7}}) z_{\max} \approx 0.26 z_{\max}.$$

It corresponds to the pressure:

Ответ:

$$P_{\max} \approx 0.093 \rho g z_{\max}$$

D1 0.40 Write down the equilibrium condition for the layer. The equation may include the pressures $P(z)$ and $P(z + \mathrm{d}z)$, $\mathrm{d}z$, the average pressure on the rod and vessel walls $\sigma(z)$, the mass of the layer $\mathrm{d}m$, as well as $\mu_1$, $\mu_2$, $g$, $a$, and $R$.

Equilibrium condition in the projection on the vertical axis (here the friction force acting on the granules from the side of the vessel is directed downwards, since the force acting on the vessel lifts the vessel):

Ответ:

$$(P(z+\mathrm{d}z) - P(z))\pi(R^2 - a^2) - \mathrm{d}mg - \mu_1 \sigma(z) \cdot 2\pi R \mathrm{d}z + \mu_2 \sigma(z) \cdot 2\pi a \mathrm{d}z = 0$$

D2 0.30 Show that the bottom pressure $P$ as a function of the fill height $h$ of the granular column can be written as:

$$P = A + Be^{-h/\lambda}.$$

Determine the parameters $A$, $B$, and $\lambda$. Express the answers in terms of the bulk density of the granules $\rho$, as well as $g$, $a$, $R$, $\mu_1$, $\mu_2$, and $K$.

The resulting equation is perfectly analogous to the equation from **B1**. Solving it, we obtain:

Ответ:

$$A = \frac{\rho g(R^2 - a^2)}{2(-\mu_1 R + \mu_2 a)K}, \quad B = -\frac{\rho g(R^2 - a^2)}{2(-\mu_1 R + \mu_2 a)K}, \quad \lambda = \frac{R^2 - a^2}{2(-\mu_1 R + \mu_2 a)K}$$

D3 0.80 Determine the minimum level $H$ of the granules relative to the bottom at which the cylinder can be lifted as described above. Assume that $H \ll \lambda$. Express the answer in terms of $a$, $R$, $\mu_2$, and $K$.
*Hint:* You may use the following approximation:

$$e^x \approx 1 + x + \frac{x^2}{2}, \quad x \ll 1$$

The dependence of the pressure $P(z)$ for a fixed level $H$ of the granules is similarly given by the formula:

$$P(z) = A + Be^{-z/\lambda}$$

Let us calculate the total force acting on the rod:

$$F = 2\pi a K \mu_2 \int_0^H P(z)\mathrm{d}z = 2\pi a K \mu_2 A(H - \lambda(1 - e^{-H/\lambda})) \approx 2\pi a K \mu_2 A \frac{H^2}{2\lambda},$$

since $H \ll \lambda$. According to Newton's third law, the same buoyant force acts on the granules. In the critical case, it must balance the gravitational force acting on the granules:

$$\rho g \pi (R^2 - a^2) H = \pi a K \mu_2 A \frac{H^2}{\lambda}$$

From this, we express $H$:

Ответ:

$$H = \frac{R^2 - a^2}{Ka\mu_2}$$

E1 0.20 Derive an equation relating $S_1$, $S_2$, $\Delta l_1$, and $\Delta l_2$.

From the incompressibility condition:

Ответ:

$$S_1 \Delta l_1 = S_2 \Delta l_2$$

E2 0.10 Determine the total energy $E$ (the sum of kinetic and potential energies) of the selected fluid volume $S_1 \Delta l_1$. Express the answer in terms of $\rho$, $S_1$, $\Delta l_1$, $v_1$, $g$, and $h_1$. Take the potential energy at $h = 0$ as the zero reference level.

Ответ:

$$W_1 = \rho S_1 \Delta l_1 \left( \frac{v_1^2}{2} + g h_1 \right)$$

E3 0.30 Determine the work $A$ done by pressure forces on the fluid bounded by the chosen stream tube during time $\Delta t = \dfrac{\Delta l_1}{v_1}$. Express the answer in terms of $p_1$, $p_2$, $S_1$, and $\Delta l_1$.

A volume $V_1$ of fluid is "pushed into" the stream tube at pressure $p_1$, and a volume $V_2 = V_1$ is "pushed out" of it at pressure $p_2$. Hence:

Ответ:

$$A = p_1 V_1 - p_2 V_2 = (p_1 - p_2) S_1 \Delta l_1$$

E4 0.30 Using the law of conservation of energy, express the pressure $p_2$ in terms of $p_1$, $\rho$, $g$, $h_1$, $h_2$, $v_1$, and $v_2$. The resulting equation is known as the **Bernoulli equation**.

We write the law of conservation of energy as:

$$\rho S_1 \Delta l_1 \frac{v_2^2 + 2gh_2 - v_1^2 - 2gh_1}{2} = (p_1 - p_2) S_1 \Delta l_1$$

By canceling $S_1 \Delta l_1$ from both sides, we obtain:

Ответ:

$$p_2 = p_1 - \frac{v_2^2 + 2gh_2 - v_1^2 - 2gh_1}{2}$$

E5 0.30 Determine the fluid outflow velocity $v(h)$ as a function of the height above the orifice. Express the answer in terms of $g$ and $h$.

From the Bernoulli equation derived in the previous section, for ideal fluid flow, the following holds:

$$p + \rho g h + \rho \frac{v^2}{2} = \text{const}$$

The pressure at the orifice is equal to the atmospheric pressure, as is the pressure at the upper boundary of the fluid and air. Thus, we have:

$$p_0 + \rho g h + 0 = p_0 + 0 + \rho \frac{v(h)^2}{2}$$

From this, it follows:

Ответ:

$$v(h) = \sqrt{2gh}$$

E6 0.60 Determine the time $\tau$ required for all the fluid initially above the orifice to drain through it. Express the answer in terms of $g$, $H$, $R$, and $D$.

Volumetric flow rate through the orifice:

$$\dot{V} = -\frac{\pi D^2}{4} \sqrt{2gh}$$

From the continuity equation, we obtain a differential equation for the water height above the orifice:

$$\pi R^2 \dot{h} = -\frac{\pi D^2}{4} \sqrt{2gh}$$

Note that since $R \gg D$, it follows that $\dot{h} \ll \sqrt{2gh}$; therefore, the assumption that the velocity far from the orifice is negligible is valid.
We can now express the drainage time:

Ответ:

$$\tau = \frac{2\sqrt{2}R^2}{D^2\sqrt{g}} \int_0^H \frac{\mathrm{d}h}{\sqrt{h}} = \frac{4\sqrt{2}R^2\sqrt{H}}{D^2\sqrt{g}}$$

F1 0.20 Due to internal friction between the granules, the maximum angle at which the free surface of the granular material can deviate from the horizontal is $\theta_r$. What is the maximum wall thickness $\omega_{\max}$ for which the flow of granules from the vessel is possible? Express the answer in terms of $D$ and $\theta_r$.

In the critical case, the material is tilted from the horizontal by an angle $\theta_r$ and is positioned at the outer edge of the orifice. From this, we obtain:

Ответ:

$$\omega_{\max} = D \operatorname{ctg} \theta_r$$

F2 0.80 Determine $q$ and $n$.

Estimate the granule velocity at the orifice exit using the law of conservation of energy:

$$v \propto \sqrt{D}$$

Estimate the mass flow rate of the granules:

$$\frac{\mathrm{d}m}{\mathrm{d}t} = \rho S v \propto D^2 \sqrt{D} \propto D^{\frac{5}{2}}$$

From this, it follows:

Ответ:

$$n = \frac{5}{2}$$

$$q = 0$$

## Решение

T **Условие**

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**Русский**

A0 [0.20] Using the given value of the radiation length of air $X_0$, calculate the splitting length $d_e$. Express your answer in g/cm$^2$ and round it to one decimal place.
**Note**: We will call the number of shower generations $n$ the quantity, which denotes how many successive splittings have occurred since the beginning of the shower development.

Ответ:

$$d_e = X_0 \ln 2 = 25.1 \text{ g/cm}^2$$

A1 [0.40] Determine the number of shower generations $n_{\max}$ after which the energy of the particles in the shower becomes less than the critical energy.

Since the energy of the shower particles is halved after each splitting length, let us determine after how many splittings the particle energy becomes smaller than the critical energy:

$$\log_2 \frac{E_\gamma}{E_e^{\text{crit}}} = 6.8 \Rightarrow$$

Ответ:

$$n_{\max} = 7$$

A2 [0.40] Determine the geometric distance $l_e$ traveled by the shower (in m) in air by the time the maximum number of particles is reached.

Let us calculate the radiation path length traversed by the shower by the time it reaches the 7th splitting:

$$x_e = d_e \cdot n_{\max} = 175.7 \text{g/cm}^2.$$

For an atmosphere of constant density $\rho_0$, this radiation path length corresponds to the following geometric distance:

$$l_e = \frac{x_e}{\rho_0} = 1.4 \text{ km}$$

A3 [0.20] Determine the maximum number of particles $N_{\max}$ in the shower.

At each splitting, the number of shower particles doubles. Therefore,

Ответ:

$$N_{\max} = 2^{n_{\max}} = 128$$

A4 [0.80] Determine the total charge $Q_e$ (in C) carried by the electrons in the shower by the time its development ceases.

At each splitting, a photon converts into an electron–positron pair and disappears, while each electron and positron emits one photon. Accordingly, the number of photons after a splitting is equal to the total number of electrons and positrons before the splitting. The number of positrons and electrons after a splitting increases by the number of photons before the splitting. Let us construct a table showing the numbers of each type of particle as a function of the number of generations.

| $n$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| $\gamma$ | 1 | 0 | 2 | 2 | 6 | 10 | 22 | 42 |
| $e^-$ | 0 | 1 | 1 | 3 | 5 | 11 | 21 | 43 |
| $e^+$ | 0 | 1 | 1 | 3 | 5 | 11 | 21 | 43 |

We find that when the shower ceases to develop, the number of electrons in it is $n_{e^-} = 43$.
Therefore, the total charge carried by the electrons is

Ответ:

$$Q_e = n_{e^-} \cdot q_e = -6.9 \cdot 10^{-18} \text{ C}$$

B1 [0.40] Determine the maximum number of generations of the hadronic component of the shower, $n_{\text{hadr}}^{\max}$.

The number of generations in the hadronic component of the shower is determined analogously to **A1**. However, in a hadronic shower, the particle energy decreases by a factor of 15 at each splitting:

$$\log_{15} \frac{E_p}{E_\pi^{\text{crit}}} = 2.2 \Rightarrow$$

Ответ:

$$n_{\text{hadr}}^{\max} = 3$$

B2 [1.00] Determine the maximum number of generations of the entire shower $n_{\text{EM}}^{\max}$, taking into account the electromagnetic component.

In a hadronic interaction, the particle energy decreases by a factor of 15, whereas in an electromagnetic interaction it decreases by a factor of 2. Therefore, the maximum number of generations is obtained when the shower develops electromagnetically from the very beginning, except for the first hadronic interaction, in which the proton decays and the $\pi^0$ meson splits into two photons:

$$\log_2 \frac{E_p/(15 \cdot 2)}{E_e^{\text{crit}}} = 11.4 \Rightarrow$$

Ответ:

$$n_{\text{EM}}^{\max} = 12 + 1 = 13$$

B3 [0.60] Determine the fraction of the shower energy carried away by muons and neutrinos, $\beta_{\text{miss}}$. Round your answer to two decimal places.

The energy carried away by muons and neutrinos is the energy remaining in the charged pions $\pi^\pm$ once their energies have fallen below the critical value. At each hadronic interaction, 6 out of 15 particles become neutral pions $\pi^0$, and their energy is transferred to the electromagnetic component. The remaining 9 out of 15 particles stay in the hadronic component.

Therefore, after the hadronic shower development ceases (i.e., after
$n_{\text{hadr}}^{\max} = 3$
splittings), the fraction of energy remaining in the hadronic component is

$$\beta_{\text{miss}} = \left(\frac{9}{15}\right)^{n_{\text{hadr}}^{\max}} = \left(\frac{3}{5}\right)^3 = 0.216.$$

All of these charged pions subsequently decay into muons and neutrinos, carrying away this energy. Thus,

Ответ:

$$\beta_{\text{miss}} = 0.22$$

C1 1.30 Determine the altitude $h_{\text{stop}}$ at which the shower development ceases. Express your answer in kilometers and round it to one decimal place.

One hadronic interaction reduces the particle energy by a factor of 15, which corresponds to $\log_2 15 = 3.9$ electromagnetic splittings. A reduction of the energy by a factor of 15 corresponds either to a radiation path length $d_h = 57\ \text{g/cm}^2$, or to a radiation path length
$3.9 \cdot d_e = 97.9\ \text{g/cm}^2$.
Therefore, the electromagnetic shower stops developing later (or farther away), and
$h_{\text{stop}}$ should be calculated specifically for the electromagnetic component.

Let us calculate the radiation path length traveled by the shower by the time it stops. This consists of one hadronic interaction, followed by the decay of $\pi^0$ into two photons and the remaining electromagnetic splittings:

$$\log_2 \frac{E/(15 \cdot 2)}{E_e^{\text{crit}}} = 15.2 \Rightarrow x = d_h + 16 \cdot d_e = 458.6 \text{g/cm}^2.$$

Let us now calculate at what geometric altitude the shower stops developing, taking into account the atmospheric density:

$$x = \int_{h_{\text{stop}}}^{\infty} \rho_0 e^{-h/H} dh = \rho_0 H e^{-h_{\text{stop}}/H}.$$

Ответ:

$$h_{\text{stop}} = H \ln \frac{\rho_0 H}{x} = 6.6\ \text{km}$$

C2 0.50 Determine what fraction of the shower energy, $\beta_{\text{EM}}$, ends up in the electromagnetic component. Round your answer to two decimal places.

By analogy with **B3**, let us find the energy carried away by muons and neutrinos:

$$\log_{15} \frac{E}{E_\pi^{\text{crit}}} = 3.14 \Rightarrow n_h = 4.$$

Then all the remaining energy stays in the EM component:

Ответ:

$$\beta_{\text{EM}} = 1 - \left(\frac{9}{15}\right)^4 = 0.87$$

C3 0.20 Estimate the shower radius $R_{\text{stop}}$ at the altitude $h_{\text{stop}}$. Express your answer in meters and round it to the nearest integer.

Let us calculate the air density at the altitude $h_{\text{stop}}$:

$$\rho(h_{\text{stop}}) = \rho_0 e^{-h_{\text{stop}}/H} = 0.57 \text{kg/m}^3.$$

Substituting this value into the radius formula, we obtain:

Ответ:

$$R_{\text{stop}} = \frac{X_0}{\rho_0 e^{-h_{\text{stop}}/H}} \cdot \frac{E_s}{E_e^{\text{crit}}} = 150\ \text{m}$$

C4 0.20 Estimate the area of the shower $S$ at the Earth's surface. Express your answer in $\text{m}^2$ and keep 2 significant figures.

Scaling the shower radius down to the Earth's surface while accounting for the change in air density, we obtain

$$R = R_{\text{stop}} \frac{\rho}{\rho_0} = 67\text{m}.$$

Ответ:

$$S = \pi R^2 = 14 \cdot 10^3\ \text{m}^2$$

C5 0.80 Estimate the characteristic signal duration $\Delta t$ at a detector located on the Earth's surface at the center of the shower. Express your answer in ns and round it to one decimal place.

Since the particles diverge at an angle $\theta = R_{\text{stop}}/h_{\text{stop}}$ from the altitude $h_{\text{stop}}$, the shower front will have the following path-length difference by the time it reaches the Earth:

$$\Delta = h_{\text{stop}} \cdot (1 - \cos\theta) \approx h_{\text{stop}} \frac{\theta^2}{2} = \frac{R_{\text{stop}}^2}{2h_{\text{stop}}} = 1.7\text{m}.$$

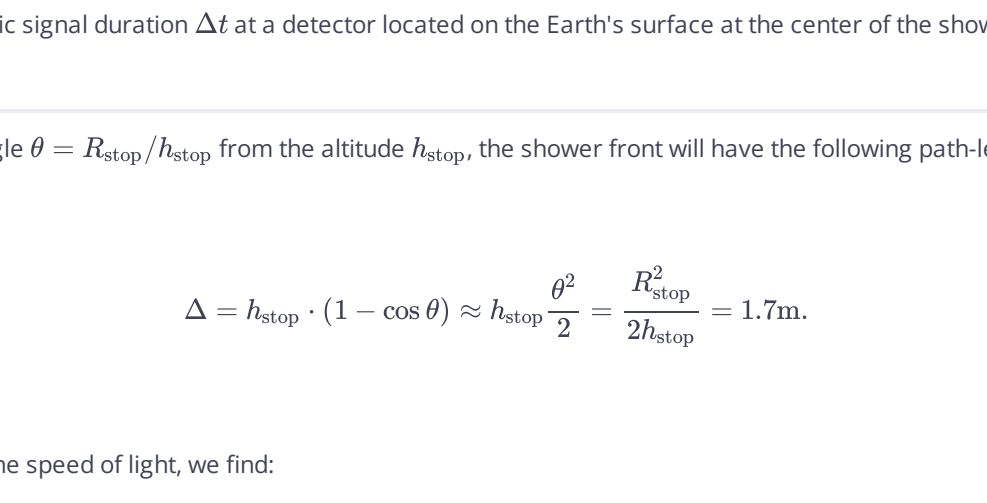
Since the particles travel nearly at the speed of light, we find:

Ответ:

$$\Delta t = \frac{\Delta}{c} = \frac{R_{\text{stop}}^2}{2ch_{\text{stop}}} = 5.7\ \text{ns}$$

D1 1.00 Determine the coefficient $\kappa$ of the detector at the Earth's surface.

To determine $\kappa$ from the data, let us construct a histogram of the detector signal:




The energy of a particle arriving at the detector cannot exceed the critical energy $E_e^{\text{crit}} = 88\ \text{MeV}$. In the histogram, one can see a sharp cutoff at the signal value $A^{\text{crit}} = 1.32 \cdot 10^8$, from which we find:

Ответ:

$$\kappa = \frac{A^{\text{крит}}}{E_e^{\text{крит}}} = 1.5 \cdot 10^9\ \text{GeV}^{-1}$$

D2 0.50 Determine the energy of the initial photon $E_0$.

Let us sum all the signal recorded by the detector:

$$A_0 = 70240347424,$$

from which, taking into account the coefficient $\kappa$, we obtain:

Ответ:

$$E_0 = \frac{A_0}{\kappa} = 46.8 \text{ GeV}$$

D3 1.50 Calculate the atmospheric density at the Earth's surface $\rho_{\text{atm}}$.

Let us calculate the mean shower radius from the data using the formula from **C3**:

$$R = \sum_i \frac{\sqrt{x_i^2 + y_i^2}}{N} = 68.3\text{m}.$$

After the shower development ceased, its radius was, according to formula (8),

$$R_{\text{stop}} = \frac{X_0}{\rho(h_{\text{stop}})} \cdot \frac{E_s}{E_e^{\text{crit}}},$$

and then it decreased inversely proportional to the density according to formula (9):

$$R = R_{\text{stop}} \frac{\rho(h_{\text{stop}})}{\rho_{\text{atm}}} = \frac{X_0}{\rho_{\text{atm}}} \cdot \frac{E_s}{E_e^{\text{crit}}}.$$

Ответ:

$$\rho_{\text{atm}} = \frac{X_0}{R} \cdot \frac{E_s}{E_e^{\text{крит}}} = 1.26 \text{ kg/m}^3$$

## Решение

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Русский

A1 0.50 Total electric charge near the junction must be equal to zero. Using this fact, write an equation connecting widths of depletion layers $d_n$, $d_p$ and impurities concentrations $N_a$, $N_d$.

Let $A$ denote the contact area. Since the volume charge density is $\rho = \pm q_e N_{a,d}$ ($+$ for holes, $-$ for electrons), the vanishing of the total hole and electron charge implies $AN_a d_p = AN_d d_n$, which yields

Ответ:

$$N_a d_p = N_d d_n.$$

A2 0.50 Show that the following equation holds for electric field

$$\frac{dE_x}{dx} = \frac{\rho}{\varepsilon_0 \varepsilon},$$

where $\rho$ – volume charge density in the corresponding point.

According to the Gauss law for electric fields in the differential form,

$$\operatorname{div} \vec{E} = \frac{\rho}{\varepsilon_0 \varepsilon}.$$

Since $\vec{E} = \vec{E}(x)$, we have

$$\operatorname{div} \vec{E} = \frac{dE_x}{dx},$$

which proves the required statement.

A3 1.00 Find the electric field $E_x(x)$ as a function of $x$. Assume that far from junction the field is equal to 0. Express the answer in terms of $N_a$, $N_d$, $d_n$, $d_p$, $\varepsilon$, $\varepsilon_0$ and $q_e$.

Using the result of part A2, as well as the relationship between concentration and volume charge density $\rho = \pm q_e N_{a,d}$ ($+$ for holes, $-$ for electrons), we get

$$\frac{dE_x}{dx} = \begin{cases} -\frac{q_e N_a}{\varepsilon_0 \varepsilon}, & -d_p \le x \le 0, \\ \frac{q_e N_d}{\varepsilon_0 \varepsilon}, & 0 < x \le d_n, \\ 0, & x < -d_p \text{ or } x > d_n. \end{cases}$$

It remains to take the indefinite integral and find the constant of integration. To obtain the answer faster, we will directly evaluate the definite integral. Since $E_x(\pm\infty) = 0$, we have

$$E_x(x) = \int_{-\infty}^{x} E'(x)\,dx = -\int_{x}^{+\infty} E'(x)\,dx.$$

We use these relations to get the final answer.

Ответ:

$$E_x(x) = \begin{cases} -\frac{q_e N_a (x + d_p)}{\varepsilon_0 \varepsilon}, & -d_p \le x \le 0, \\ \frac{q_e N_d (x - d_n)}{\varepsilon_0 \varepsilon}, & 0 < x \le d_n, \\ 0, & x < -d_p \text{ or } x > d_n. \end{cases}$$

A4 1.00 Find the electrostatic potential $\varphi(x)$ as a function of $x$. Assume that $\varphi(0) = 0$. Express the answer in terms of $N_a$, $N_d$, $d_n$, $d_p$, $\varepsilon$, $\varepsilon_0$ and $q_e$. Determine the potential difference at large distance to the junction $\Delta\varphi = \varphi(x = +\infty) - \varphi(x = -\infty)$.

In the one-dimensional case, $\varphi'(x) = -E_x(x)$. Using the condition $\varphi(0) = 0$, we can express the electrostatic potential as:

$$\varphi(x) = -\int_0^x E_x(x)\mathrm{d}x.$$

Evaluating the integral explicitly, we obtain:

Ответ:

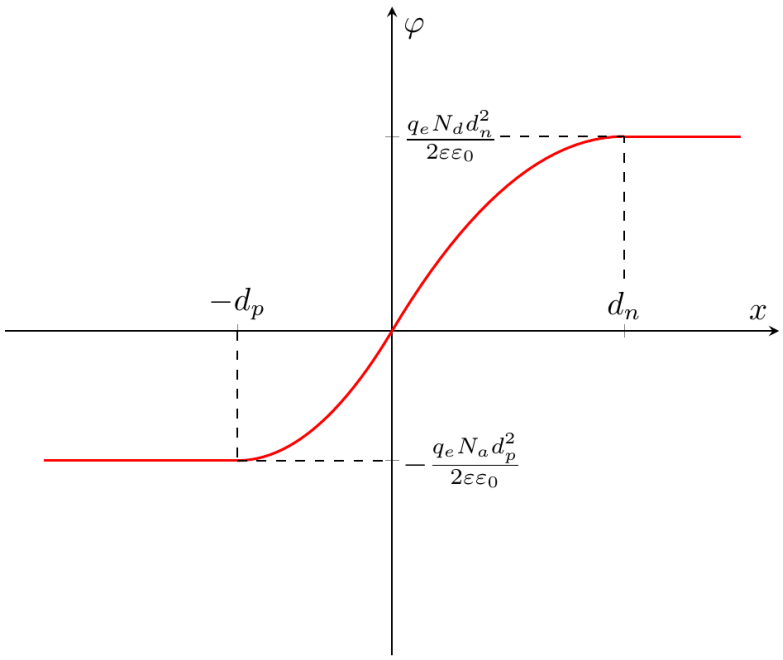
$$\varphi(x) = \begin{cases} -\frac{q_e N_a d_p^2}{2\varepsilon_0\varepsilon}, & x < -d_p, \\ \frac{q_e N_a (x^2 + 2d_p x)}{2\varepsilon_0\varepsilon}, & -d_p \le x < 0, \\ \frac{q_e N_d (2d_n x - x^2)}{2\varepsilon_0\varepsilon}, & 0 \le x < d_n, \\ \frac{q_e N_d d_n^2}{2\varepsilon_0\varepsilon}, & x \ge d_n. \end{cases}$$

Ответ:

$$\Delta\varphi = \frac{q_e(N_a d_p^2 + N_d d_n^2)}{2\varepsilon\varepsilon_0}.$$

A5 0.50 Draw a qualitative plot of the potential $\varphi(x)$ as function of $x$.

Ответ:




A6 1.00 Actually, the potential difference is determined by the material properties and the temperature. Knowing the potential difference, one can calculate depletion layers widths. Express the widths $d_n$, $d_p$ in terms of $\Delta\varphi$, $N_a$, $N_d$, $\varepsilon$, $\varepsilon_0$ and $q_e$. Calculate their numerical values for $\Delta\varphi = 0.8$ V.

From part A4 we have

$$\Delta\varphi = \frac{q_e(N_a d_p^2 + N_d d_n^2)}{2\varepsilon\varepsilon_0}.$$

Using the equation $N_a d_p = N_d d_n$ obtained in part A1 we get

$$\Delta\varphi = \frac{q_e\left(N_a d_p^2 + \frac{N_a^2}{N_d} d_p^2\right)}{2\varepsilon\varepsilon_0} = \frac{q_e N_a (N_a + N_d) d_p^2}{2\varepsilon\varepsilon_0 N_d}.$$

Similarly we have

$$\Delta\varphi = \frac{q_e\left(\frac{N_d^2}{N_a} d_n^2 + N_d d_n^2\right)}{2\varepsilon\varepsilon_0} = \frac{q_e N_d (N_a + N_d) d_n^2}{2\varepsilon\varepsilon_0 N_a}.$$

what implies the answer:

Ответ:

$$d_p = \sqrt{\frac{2\varepsilon\varepsilon_0 \Delta\varphi N_d}{q_e N_a (N_a + N_d)}} \approx 58.7 \text{ nm},$$

$$d_n = \sqrt{\frac{2\varepsilon\varepsilon_0 \Delta\varphi N_a}{q_e N_d (N_a + N_d)}} \approx 117.5 \text{ nm}.$$

A7 0.50 Determine the maximum value of the electric field in the junction $E_{\max}$. Express the answer in terms of $\Delta\varphi$, $N_a$, $N_d$, $\varepsilon$, $\varepsilon_0$ and $q_e$. Calculate its numerical value for $\Delta\varphi = 0.8$ V.

It follows from the $E_x(x)$ dependence obtained in part A3 that the maximal absolute value of the electric field is achieved at $x = 0$ and equals

$$E_{\max} = \frac{q_e N_a d_p}{\varepsilon\varepsilon_0}.$$

Substituting the expression for $d_p$ obtained in part A6 we get

Ответ:

$$E_{\max} = \sqrt{\frac{2 q_e \Delta\varphi N_a N_d}{\varepsilon\varepsilon_0 (N_a + N_d)}} \approx 9.08 \cdot 10^6 \text{ V/m}.$$

B1 0.50 Let $n_{10}$ denote the concentration of electrons in $p$-type semiconductor without external voltage. Similarly let $p_{10}$ denote the concentration of holes in $n$-type semiconductor without external voltage. Determine electron concentration $n_1$ in $p$-type semiconductor and hole concentration $p_1$ in $n$-type semiconductor after the voltage $V$ is applied. Consider region outside of depletion layers, where electric field is zero. Express the answers in terms of $p_{10}$, $n_{10}$, $V$, $T$, $k$ and $q_e$.

It follows from the formula for the distribution of $n(x)$ that

$$\frac{n(x = +\infty)}{n(x = -\infty)} = \exp\left(-\frac{U_n(x = +\infty) - U_n(x = -\infty)}{kT}\right)$$

Before the voltage is applied, we have $n(x = +\infty) = N_d$, $n(x = -\infty) = n_{10}$. After the voltage is turned on, the difference of potential energies $U_n(x = +\infty) - U_n(x = -\infty)$ for a single electron changes by $(-q_e) \cdot (-V) = q_e V$. According to the problem text, the electron concentration in the $n$-type semiconductor equals $N_d$ for any voltage. So after applying the voltage we still have $n(x = +\infty) = N_d$, but $n(x = -\infty) = n_1$. Thus we get

$$\frac{N_d}{n_1} = \exp\left(-\frac{q_e V}{kT}\right) \cdot \frac{N_d}{n_{10}},$$

which implies

$$n_1 = n_{10} \cdot \exp\left(\frac{q_e V}{kT}\right).$$

The expression for $p_1$ can be obtained similarly. It follows from the formula for the distribution of $p(x)$ that

$$\frac{p(x = +\infty)}{p(x = -\infty)} = \exp\left(-\frac{U_p(x = +\infty) - U_p(x = -\infty)}{kT}\right).$$

Before the voltage is applied, we have $p(x = +\infty) = p_{10}$, $p(x = -\infty) = N_a$. After the voltage is turned on, the difference of potential energies $U_p(x = +\infty) - U_p(x = -\infty)$ for a single hole changes by $q_e \cdot (-V) = -q_e V$. According to the problem text, the concentration of holes in the $p$-type semiconductor equals $N_a$, so that after applying the voltage we still have $p(x = -\infty) = N_p$, but $p(x = +\infty) = p_1$. Thus we get

$$\frac{p_1}{N_a} = \exp\left(\frac{q_e V}{kT}\right) \cdot \frac{p_{10}}{N_a},$$

which implies

$$p_1 = p_{10} \cdot \exp\left(\frac{q_e V}{kT}\right).$$

Ответ:

$$n_1 = n_{10} \cdot \exp\left(\frac{q_e V}{kT}\right), \quad p_1 = p_{10} \cdot \exp\left(\frac{q_e V}{kT}\right).$$

B2 1.00 Consider the volume between the planes with coordinates $x$ and $x + \Delta x$ (Fig. 3). Write down the condition that the holes concentration in this volume does not change. Take into account the holes flux through the two planes and holes recombination. In the limit $\Delta x \to 0$ obtain a differential equation for the concentration deviation from equilibrium value $\Delta p = p_1 - p_{10}$ as a function of $x$. The answer may also include $D_p$ and $\tau_p$.

Let $A$ denote the cross section area of the $n$-type semiconductor. Consider a small time interval $\Delta t$ and calculate the change number of holes change within the small volume under consideration over this time interval. The change due to recombination is $\left(\frac{dp_1}{dt}\right)_{\text{rec}} A\Delta x \Delta t$. The changes due to diffusion through the two planes are $J_p(x)A\Delta t$ (at coordinate $x$) and $-J_p(x+\Delta x)A\Delta t$ (at coordinate $x + \Delta x$). The total number of holes, and therefore concentration, are constant, so the total change must be zero:

$$\left(\frac{dp_1}{dt}\right)_{\text{rec}} A\Delta x \Delta t + J_p(x)A\Delta t - J_p(x+\Delta x)A\Delta t = 0.$$

Thus we get

$$\left(\frac{dp_1}{dt}\right)_{\text{rec}} = \frac{J_p(x+\Delta x) - J_p(x)}{\Delta x} = \frac{dJ_p(x)}{dx} = -D_p \frac{d^2 p_1}{dx^2}.$$

Since $\Delta p$ and $p_1$ differ by a constant,

$$\frac{d^2 p_1}{dx^2} = \frac{d^2 \Delta p}{dx^2}.$$

Taking into account the equation

$$\left(\frac{dp_1}{dt}\right)_{\text{rec}} = -\frac{\Delta p}{\tau_p}$$

we obtain:

Ответ:

$$\frac{d^2 \Delta p}{dx^2} = \frac{\Delta p}{\tau_p D_p}$$

B3 0.60 Find the distance $L_p$, over which the deviation of holes concentration from equilibrium decreases by $e$ times. Express the answer in terms of $D_p$, $\tau_p$. Compute numerical value for $D_p = 1.0 \cdot 10^{-3}$ m$^2$/s, $\tau_p = 5.0 \cdot 10^{-7}$ s.

The general solution of the equation from part B2 has the following form:

$$\Delta p = a \exp\left(-\frac{x}{\sqrt{\tau_p D_p}}\right) + b \exp\left(\frac{x}{\sqrt{\tau_p D_p}}\right)$$

for some coefficients $a$ and $b$. As $x \to +\infty$ the hole concentration must approach the equilibrium value $p_{10}$, so that $\Delta p \to 0$, and thus $b = 0$. Therefore,

$$\Delta p \sim \exp\left(-\frac{x}{\sqrt{\tau_p D_p}}\right),$$

and hence $\Delta p$ changes by a factor of $e$ when $x$ changes by

Ответ:

$$L_p = \sqrt{\tau_p D_p} \approx 2.24 \cdot 10^{-5} \text{ m}.$$

B4 0.80 Determine the hole concentration $p_1(x)$ in region $x > d_n$ as a function of coordinate (in $n$-type semiconductor outside of the depletion layer). Express the answer in terms of $p_{10}$, $V$, $T$, $L_p$, $d_n$, $k$ and $q_e$.

From the expression for $\Delta p$ obtained in part B3, we find for $x > d_n$:

$$p_1(x) = p_{10} + a\exp\left(-\frac{x}{L_p}\right),$$

where $a$ is some coefficient. Within the depletion layer, the concentration $p(x)$ obeys the Boltzmann distribution. Therefore,

$$\frac{p(d_n)}{p(-d_p)} = \exp\left(-\frac{U_p(d_n) - U_p(-d_p)}{kT}\right).$$

According to the problem text, the electric field outside the depletion layer can be assumed to be zero, so the voltage change falls entirely within the region $-d_p \le x \le d_n$. Therefore, when the voltage is applied, the potential energy difference $U_p(d_n) - U_p(-d_p)$ for a single hole changes by $-q_eV$, so that similarly to part B1 we can write

$$\frac{p_1(d_n)}{N_a} = \exp\left(\frac{q_eV}{kT}\right)\frac{p_{10}}{N_a},$$

which implies

$$p_1(d_n) = \exp\left(\frac{q_eV}{kT}\right)p_{10} = p_{10} + a\exp\left(-\frac{d_n}{L_p}\right),$$

so that

$$a = p_{10}\left(\exp\left(\frac{q_eV}{kT}\right) - 1\right)\exp\left(\frac{d_n}{L_p}\right).$$

Ответ:

$$p_1(x) = p_{10} + p_{10}\left(\exp\left(\frac{q_eV}{kT}\right) - 1\right)\exp\left(-\frac{x - d_n}{L_p}\right)$$

B5 0.80 Determine the density of hole diffusion current $j_p$ at the boundary of depletion layer $x = d_n$. Express the answer in terms of $p_{10}$, $V$, $L_p$, $D_p$, $T$, $k$ and $q_e$.

To calculate the current of holes, let us consider the $n$-type region near the boundary of the depletion layer, where the flux of holes is determined by diffusion:

$$J_p = -D_p\frac{dp_1}{dx} = \frac{D_pp_{10}}{L_p}\left(\exp\left(\frac{q_eV}{kT}\right) - 1\right)\exp\left(-\frac{x - d_n}{L_p}\right)\Bigg|_{x=d_n} = \\ = \frac{D_pp_{10}}{L_p}\left(\exp\left(\frac{q_eV}{kT}\right) - 1\right).$$

The current density is $j_p = q_eJ_p$, which yields

Ответ:

$$j_p = \frac{q_eD_pp_{10}}{L_p}\left(\exp\left(\frac{q_eV}{kT}\right) - 1\right).$$

B6 0.80 The junction cross section area is $A$. Derive the expression for the total current $I$ through the junction as a function of applied voltage $V$, semiconductor parameters $N_a$, $N_d$, $\tau_p$, $\tau_n$, $D_p$, $D_n$, $n_i$ and constants $k$, $q_e$.

Similarly to part B5, the expression for the electron current has the form

$$j_n = \frac{q_eD_nn_{10}}{L_n}\left(\exp\left(\frac{q_eV}{kT}\right) - 1\right).$$

Taking into account that

$$p_{10} = \frac{n_i^2}{N_d}, \quad n_{10} = \frac{n_i^2}{N_a},$$

и we find the required current as the sum of the hole and electron currents:

$$I = q_e A n_i^2 \left( \exp\left( \frac{q_e V}{kT} \right) - 1 \right) \left( \frac{D_p}{L_p N_d} + \frac{D_n}{L_n N_a} \right).$$

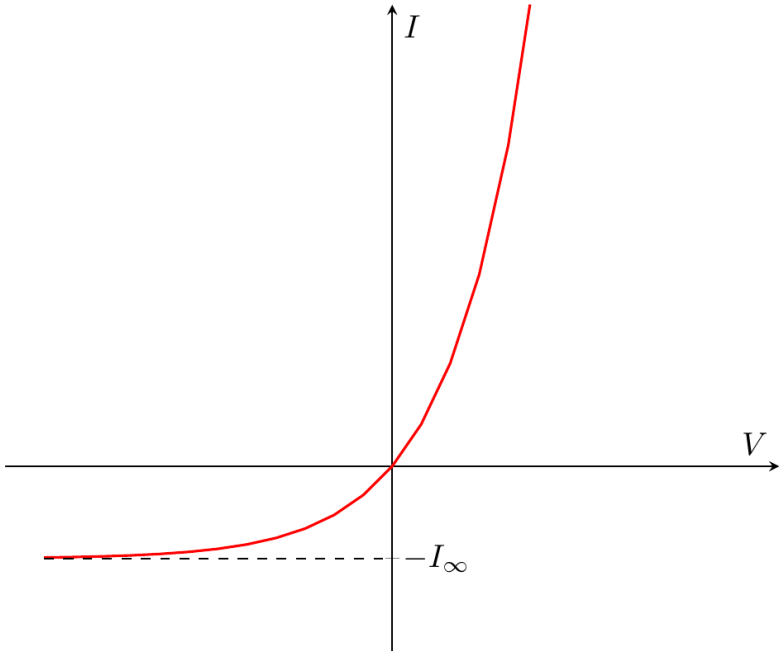
Using the answer in part B3, we obtain:

Ответ:

$$I = q_e A n_i^2 \left( \exp\left( \frac{q_e V}{kT} \right) - 1 \right) \left( \frac{1}{N_d} \sqrt{\frac{D_p}{\tau_p}} + \frac{1}{N_a} \sqrt{\frac{D_n}{\tau_n}} \right).$$

B7 0.30 Draw a qualitative plot of the relation $I(V)$, obtained in previous task.

Ответ:




B8 0.20 Calculate the numerical values of the saturation current $I_\infty$ at large negative voltage, and the value of current $I_1$ for $V = 0.7$ V.

The saturation current is

$$I_\infty = q_e A n_i^2 \left( \frac{1}{N_d} \sqrt{\frac{D_p}{\tau_p}} + \frac{1}{N_a} \sqrt{\frac{D_n}{\tau_n}} \right) = 5.77 \cdot 10^{-16} \text{ A},$$

and the current at $V = 0.7$ V equals to $I_1 = 3.23 \cdot 10^{-4}$ A.

Ответ:

$$I_\infty = 5.77 \cdot 10^{-16} \text{ A}, \quad I_1 = 3.23 \cdot 10^{-4} \text{ A}.$$